

1. Current Attacks

In April, the Play ransomware group exploited a zero-day vulnerability in the Windows Common Log File System (CLFS), known as CVE-2025-29824. This vulnerability allowed local attackers to escalate privileges to system level. Using this flaw, the attackers deployed PipeMagic, a loader that enabled further execution of Play ransomware and related malware across the network.

Zero-day Exploit (CVE-2025-29824)

A use-after-free flaw in `clfs.sys` allowed creation of a malformed CLFS handle, triggering privilege escalation.

Optimization with `dllhost.exe` enabled SYSTEM privileges. This bypassed typical protections like standard user privileges, preventing detection by URL filters or network security appliances.

The campaign included running data theft tools, then preparing for ransomware deployment with data encryption and potential leak threats, typical of Play's tactics.

Could the attack have been prevented?

Yes, the attack could have been prevented or its impact significantly reduced:

- Reducing service and process privileges, and isolating critical system components, could have limited SYSTEM-level attacks even post-exploitation.
- Blocking lateral movement between network segments and monitoring PowerShell and MSBuild activity could have detected early-stage reconnaissance and halted the spread.
- Alerts on CLFS exploitation, unusual credential-dumping activity, or use of certutil could have allowed response teams to intervene before ransomware deployment.

Sources:

- [Play ransomware exploited Windows logging flaw in zero-day attacks](#)
- [Ransomware Landscape May 2025: SafePay, DevMan Emerge as Major Threats](#)
- [Play \(Hacker group\)](#)

2. Malware

2.1. How many new malware variants are reported daily? Name your source with report name, year, and page number.

According to AV-TEST, over 450,000 new malicious programs or PUAs are registered worldwide each day. The global average is around 190,000 new malware attacks every second, with nearly 90% being phishing and social-engineering-based [avg.com](#).

Source: [AV-TEST Institute - Malware Statistics & Trends Report, 2025, p. 1](#)

2.2. What is a botnet? What is the most common operating system targeted by botnets? Name your source with report name, year, and page number.

Botnet Example: Mirai

- Predominantly targets Linux-based IoT devices (e.g. IP cameras, routers and etc.) ([Mirai \(malware\) - Wikipedia](#)).
- Used mainly for DDoS attacks, but also to infiltrate networks or deploy droppers.

Germany was the most targeted country for DDoS in Q1 2025. Over 20.5 million attacks occurred, with techniques like Mirai botnet amplifications commonly used ([Germany Most Targeted Country in Q1 2025 DDoS Attacks](#)).

2.3. What are typical attacker actions before executing ransomware? Name your source with report name, year, and page number.

Based on frameworks like the *BSI kill-chain model* and *Unit 42 Q1 2025* report:

Initial Access:

- Phishing emails with malicious attachments or links; sometimes exploiting remote access flaws like unpatched RDP [\(Top 10 Ransomware measures\)](#).

Exploit and Privilege Escalation:

- Zero-day exploits e.g. CLFS vulnerability or credential-dumping tools .

Lateral Movement and Credential Dumping:

- Tools such as Mimikatz target LSASS exploring internal networks .

Data Exfiltration (Double Extortion):

- Sensitive data stolen before encryption and threatened to be made public [\(Ransomware\)](#).

Encryption and Ransom Demand:

- Common file extensions: `.encrypted` , `.play` , `.lock` and etc.
- Ransom demanded in Bitcoin or mobile top-ups [\(Ransomware - Wikipedia\)](#).

2.4. What protection mechanisms are recommended by IT-Grundschutz against malware? Name the title and ID of the specific requirements and state whether these are required for basic protection, standard protection, or for increased protection needs.

According to the BSI's [Top 10 Ransomware measures](#), the following align with IT-Grundschutz modules OPS and SYS:

BSI Measure	Relevant IT Grundschutz Domain
1. Patches and Updates	OPS, SYS: Patch management, update roll-out
2. Remote Access	OPS: VPN + 2FA
3. No macros / phishing filters	OPS: Mail configuration, user training
4. Software Whitelisting	SYS: Application control
5. Antivirus + IPS	OPS, SYS: EDR/AV installations
6. Admin account segregation	SYS: Principle of least privilege
7. Network Segmentation	SYS, OPS: Firewalls, sub-networks
8. Offline and tested backups	OPS: Backup strategy, restore procedures

Additional recommendations:

- User awareness training e.g. spotting malicious links [cyfirma.com](#), [srd-rechtsanwaelte.de](#), [guidepointsecurity.com](#).
- EDR and anomaly detection to monitor CLFS activity, process spawning.
- Incident Response planning: Isolation of infected systems, documentation, notification to authorities .
- Immutable and offline backups, using app-whitelisting, ZFS snapshots or VSS hardening [en.wikipedia.org](#).

3. Vulnerability Management with CVE

The first issue i have found is CVE-2023-23923^[1]. This issue involves how someone could change another person's 'start page' in an app without needing special permissions. It seems the system didn't check thoroughly when users set their own dashboard or landing page, allowing an outsider to redirect you elsewhere. The score for this issue is 8.2, as it concerns keeping information private and preventing unauthorized changes to your settings. A workaround, if no update is available, is to limit access to preference editing through role overrides or to disable the feature using a local plugin override.

The second issue, CVE-2023-5540^[2], relates to a module called IMSCP that manages course materials. From what I understand, if someone is logged in, they could insert harmful code into their uploaded course package, which the server would execute. This issue scores 8.8 because it can disrupt server files, crash parts of the system, or leak user data, affecting privacy, data safety, and system uptime simultaneously. While waiting for an official update, the workaround is to disable the IMSCP module in the admin settings and monitor recent content uploads to ensure no one is sneaking in suspicious package files.

Lastly, there's CVE-2023-35133^[3], which consists of two problems: one where someone can trick the system into executing unexpected database queries (SQL injection) and another where they can make the server access internal resources it shouldn't (SSRF). This issue has a score of 7.5. It focuses on protecting the database from tampering and preventing outsiders from exploring the network. As a temporary solution, it is advised to turn off vulnerable services, like the MNet connection, and use firewall rules to block any unusual outgoing requests from the server until a patch is released.



1. [CVE-2023-23923](#) ↩

2. [CVE-2023-5540](#) ↩

3. [CVE-2023-35133](#) ↩